

EDUCATION	University of California San Diego (UCSD) Ph.D., Communication Theory and Systems Committee: Bhaskar D. Rao (Chair), Piya Pal, Xinyu Zhang, Philip E. Gill, Rayan Saab	2017-2023
	École polytechnique fédérale de Lausanne(EPFL) Master of Science(M.Sc.), Communication Systems	2014-2017
	Indian Institute of Technology Jodhpur(IITJ) Bachelor of Technology(B.Tech), Electrical Engineering	2010-2014
RESEARCH INTERESTS	Millimeter-wave sensing and beam alignment, Bayesian methods for sparse signal recovery, super-resolution, hybrid modeling (AI/ML/data-driven components augmented model based system design) for learning and inference, compressive/adaptive sensing	
HONOURS	2024, 2025 Long-Term Incentive Program Award, Samsung Semiconductor Inc. Rising Star in Signal Processing, ICASSP 2023 Graduate and Professional Student Association Travel Grant, UC San Diego 2023 Finalist - Qualcomm Innovation Fellowship 2020 Electrical and Computer Engineering Department Fellowship (2017), UCSD M.Sc Research Scholarship (2014-2016), EPFL Merit-cum-Means (MCM) Scholarship (2011-2014), IITJ	
PROFESSIONAL EXPERIENCE	Samsung Semiconductor, Inc., San Diego, USA <i>Modem Systems, 5G/6G Advanced Communication Systems and Signal Processing</i> <i>Staff Engineer</i> <i>Senior Engineer</i> Manager: Dr. Federico Penna	Mar.26-Present (2.5 years) Sept.23-Feb.26
	• Working on research and algorithm development for 5G/6G cellular modem • Proposed new ideas for the enhancement of both channel estimation (CE) and channel state information (CSI) blocks, validated them through link-level simulations • Participating in standardization efforts on CSI feedback and was a Samsung collaborator with Stanford University (with P.I. Prof. John M. Cioffi, Stanford University)	
	Samsung Semiconductor, Inc., San Diego, USA <i>Improving robustness of Time-Averaged Specific absorption rate (TAS) algorithm with proximity sensors</i> Supervisor: Dr. Pranav Dayal	(13 weeks) Jun.21-Sept.21
	• Designed power control algorithms to maintain SAR under regulatory norms • Proposed novel ideas to reduce the impact of sensor imperfections on TAS algorithm; contribution resulted in internal technical report	
	NOKIA Bell Labs, IL, USA <i>Enhancing 5G Coverage via LEO Satellite Links</i> Supervisor: Dr. Amitava Ghosh & Dr. Frank Hsieh	(10 weeks) Jun.19-Aug.19
	• Evaluated impact of large round trip time (RTT) on Hybrid ARQ; designed (with Dr. Hsieh) and tested new ideas for exploiting pre-NACK opportunities; performed link budget calculations and scenario set-up	
	Sony (Stuttgart Technology Center), Germany <i>Indoor Localization based on Geomagnetic Fingerprinting using MATLAB</i> Supervisor: Dr. Daniel Schneider	Feb.16-Jul.16
	• Improved the design of Step Detection algorithm for the motion model • Optimized data collection code for scalability; developed large scale magnetic fingerprints and performed statistical evaluations; improved the design of Step Detection algorithm	
	University of Manitoba, Canada <i>Improvement of RF Interrogation System Employed in High-Voltage Electric Field Measurement</i> Supervisors: Prof. Behzad Kordi and Prof. Douglas Thomson	May.13-Jul.13
	• Implemented Arduino-based code to perform RF sweep on a Phase Locked Loop	

PUBLICATIONS	R. R. Pote, B. D. Rao, "Theory and Practice of Light-Weight Sequential SBL Algorithm: An Alternative to OMP", IEEE Transactions on Signal Processing, 2025.		
	M. A. Mohsin, S. Bhattacharya, K. Rajabalifardi, R. Pote, J. M. Cioffi, "Optimum power allocation for low rank Wi-Fi channels: A comparison with deep RL framework," IEEE International Conference on Communications 2025		
	R. Pote, H. Kwon, J. M. Cioffi, D. Bai, "Optimal Power Allocation Using Shallow-Cut Approach Exploiting Time-Sharing Structure," IEEE International Conference on Communications 2025		
	R. Pote, F. Penna, H. Kwon, D. Bai, "Robust Estimation of Channel Statistics from Narrowband Reference Signals," IEEE International Conference on Communications 2025		
	S. Bhattacharya, K. Rajabalifardi, M. A. Mohsin, R. Pote, J. M. Cioffi, "Optimal Power Allocation and Time Sharing in Low Rank Multi-carrier Wi-Fi Channels," 2024 IEEE Globecom Workshops		
	R. R. Pote, B. D. Rao, "Novel Active Sensing and Inference for mmWave Beam Alignment Using Single RF Chain Systems," in arXiv, 2024 (in preparation for submission to Trans. on Wireless Communications)		
	R. R. Pote, B. D. Rao, "Novel Sensing Methodology for Initial Alignment Using MmWave Phased Arrays," 2023 Asilomar Conference on Signals, Systems and Computers, IEEE.		
	R. R. Pote, B. D. Rao, "Maximum Likelihood-based Gridless DoA Estimation Using Structured Covariance Matrix Recovery and SBL with Grid Refinement," IEEE Transactions on Signal Processing, vol. 71, pp. 802–815, 2023.		
	R. R. Pote, B. D. Rao, "Light-Weight Sequential SBL Algorithm: An Alternative to OMP," 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).		
	R. R. Pote, B. D. Rao, "Maximum Likelihood Structured Covariance Matrix Estimation and connections to SBL: A Path to Gridless DoA Estimation," 2022 Asilomar Conference on Signals, Systems and Computers.		
	R. R. Pote, B. D. Rao, "A Novel Bayesian Approach for the Two-Dimensional Harmonic Retrieval Problem," 2021 IEEE ICASSP.		
	R. R. Pote, B. D. Rao, "Robustness of SBL in correlated environments," 2020 IEEE ICASSP.		
	R. R. Pote, B. D. Rao, "Reduced dimension beamspace design incorporating nested array for mmwave channel estimation," 2019 Asilomar Conference on Signals, Systems and Computers, IEEE.		
SKILLS	Simulation environment	MATLAB, GNURadio, LabVIEW, GNS3	
	SDR Platforms	BladeRF	
	Computer Languages	C, C++, Python	
RELEVANT COURSES	Information Theory, Numerical Optimization, Sparsity & Compressed Sensing, ML for Image Processing, Probabilistic Reasoning and Decision-Making, DSP I, Stochastic Processes, Linear Algebra, Convex Optimization, Big Network Data, Advanced Digital Communications, Software Defined Radio.		
SELECTED PROJECTS	Object Detection: Objects as Points		Oct.19-Dec. 19
	Identified potential shortcoming of the architecture proposed in reference; implemented new design; improved detection score for smaller objects (Code: https://github.com/rohanpote/MoDL_CenterNet)		
	(Master Thesis) K-Gaussians Coded Caching Problem		
	Supervisor - Prof. Michael Gastpar		Sept.16-Feb.17
	- Developed methods to solve the 2-Gaussians caching problem; computed an inner bound to the K-Gaussians Coded Caching Problem with circulant correlation structure - Obtained approximate numerical solution to the optimization problem over a non-convex domain for K-Gaussians and verified the conjecture in reference numerically.		
	GPS Receiver using Software Defined Radio		
	Supervisor - Prof. Bixio Rimoldi		Sept.15-Nov.15
	Received C/A GPS signal using low-cost/open-source SDR (bladeRF) and LNA to boost the signal; implemented code to decode GPS signal and computed receiver position (Accuracy ~ 15m)		
	Understanding Social Circles in Ego Networks using Python		Dec. 17
	- Detected social circles using greedy modularity optimization, a community detection algorithm - Demonstrated that network structure can capture medium size social circles by comparing with user identified circles (Dataset: https://snap.stanford.edu/data/ego-Facebook.html)		
TALKS	Efficient Techniques for Millimeter Wave Sensing and Beam Alignment, Sparse Recovery, and DoA Estimation, IIT Hyderabad		Mar. 2025
	Efficient Techniques for Millimeter Wave Sensing and Beam Alignment, Sparse Recovery,		

and DoA Estimation, IIT Bombay Nov. 2023
On Sparse Array Geometry: Applications to mmWave Channel Estimation, Center for
 Wireless Communications Research Review (video link) Nov. 2019
Cryptographic Reverse Firewalls, Students' Seminar-Security Protocols and Applications, EPFL,
 Reference: I. Mironov and N. Stephens-Davidowitz. *Cryptographic reverse firewalls*. Cryptology ePrint
 Archive, Report 2014/758, 2014. Apr. 2015

PROFESSIONAL SERVICE **Session Chair:** Asilomar conference on Signals, Systems and Computers, 2023
Reviewer: ICASSP 20(21),-22,-23; IEEE Signal Processing Letters; Machine Learning for Signal
 Processing Workshop 2021; IEEE Transactions on Aerospace and Electronic Systems, IEEE Open
 Journal of Signal Processing, IEEE Transactions on Communication, OJSP-2024
Grad Pals Program Mentor: Assisted one incoming international graduate student during the
 COVID-19 pandemic to transition in university after deferral.
2022 Summer Research Conference Moderator: Moderated an ECE undergraduate student
 researchers' panel at UCSD to ensure timely flow of presentations and engagement in Q&A

TEACHING & GRADING EXPERIENCE **Lecturer (UCSD)** ECE 251A - Digital Signal Processing I, lecture on 'Random Process Modeling
 and Power Spectral Density Estimation' Feb.23
Teaching Assistant (UCSD) ECE 251A - Digital Signal Processing I Jan.23-Mar.23
Teaching Assistant (UCSD) CSE 7 - Introduction to Programming with MATLAB Aug.18-Sept.18
 • Conducted lab sessions for 30 students with diverse majors (with another TA and tutor)
Teaching Centre for Social Responsibility & Leadership, Mumbai Jul.14-Aug.14
 • Taught Chemistry to 55 under-privileged students preparing for Joint-Entrance Exams held by IITs

OTHER ACTIVITIES **Assistant coordinator**, Counselling Service Team, IIT Jodhpur 2012-2013
 • Organised Orientation Programme'12 attended by 150 freshmen students and their parents
Participant, Technology Vision 2035, Government of India Sept.12
 • Represented my Institute and presented disruptive technologies that I believe would evolve by 2035

PERSONAL INFORMATION Nationality Indian
 Date of Birth 23 October 1992
 Hobbies Listening and learning Indian Classical (Hindustani) vocals
 Languages Marathi (Mother Tongue)
 Hindi (Native Proficiency)
 English (Native Proficiency)